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EIST 6110 Instructional Design

Instructional Design Project, Part B: Analysis

Building Course Sections in Banner: A Training Module for Schedule Builders

Needs Analysis

Problem summary: Staff who build course sections in Banner have never received formal training, and their errors account for the largest share of procedural error tickets sent to the Registrar's technical team. Part A described two related problems: the university has no Banner training program, and staff have no training on the specific actions that cause errors. This module targets the second problem by focusing on the single action that produces the most tickets: building a course section. It is the first module of the larger role-based program from Part A.

Note: figures marked (est.) are planning estimates based on ticket history and conversations with staff. They will be confirmed with a ticket report and a short survey before design begins.

Banner terms used in this analysis:

Term	Plain meaning
Banner	The university's student information system, which holds course, registration, and student records
Course section	One scheduled offering of a course in a term, such as ENGL 1101-001 in Fall
CRN	Course reference number, the 5-digit ID students use to register for a section
SSASECT	The Banner page where a section is built (term, campus, delivery method, meeting time, instructor)
SSADETL	The Banner page for extra section details, such as attributes that flag a section as online
SSAXLST	The Banner page that links sections taught together (cross-listed) so they share one enrollment limit
DegreeWorks	The degree audit system that reads Banner data to show students which requirements a course meets
Test instance	A separate practice copy of Banner with sample data, where changes do not affect real students

Comparative Needs

Question: How does the accuracy of sections built by department schedule builders compare to sections built by trained Registrar scheduling staff?

Registrar scheduling staff learned section building through years of daily use and close contact with the technical team. Department schedule builders usually build sections only during the two or three weeks of each term's schedule build. Over the past three terms, the technical team averaged 45 procedural error tickets per term. About 18 of these per term (est., 40%) trace to course section builds, and nearly all of those came from department-built sections.

Felt Needs

Questions asked of schedule builders and Registrar scheduling staff:

- What part of building a section is most confusing?
- When you are unsure how to code a section, what do you do?
- What would help you build sections correctly the first time?

Summary of responses:

- Schedule builders said they learned by copying last term's sections or by asking the person who held the job before them. Most did not know why certain fields matter.
- The most confusing areas were distance education coding (campus, instructional method, and attributes), cross-listing, and when to use section attributes on SSADETL.
- Registrar scheduling staff said they spend a large part of each schedule build fixing sections by hand. They suggested hands-on practice in the test environment and a one-page checklist.

Expressed Needs

The Registrar's technical team and scheduling staff have asked for formal section-build training for department schedule builders. The request grew out of the ticket volume described in Part A.

Anticipated Needs

Schedule builder roles turn over often because the duty is usually added to an administrative assistant's job. Each new person currently learns by trial and error against the live system. The university is also expanding online and hybrid offerings, which makes distance education coding errors more likely if nothing changes.

Critical Incident Needs

Section errors reach students. A section coded with the wrong instructional method can charge the wrong course fees and show the wrong format during registration. A cross-listed group with mismatched enrollment caps can overfill a room. A missing attribute can keep a course from counting correctly in a student's DegreeWorks audit. These problems are often found only after registration opens.

Other consequences go beyond repair time. Students may be charged the wrong fees and need refunds, register for a format they did not expect, or be blocked from a full section that should have had room. Faculty may arrive to a room that is double-booked. Bad delivery codes also feed into state and accreditation reports on distance education.

Errors are discovered in four main ways: Registrar staff catch some during schedule review, students or advisors report problems once registration opens, Student Accounts finds fee mismatches during billing, and DegreeWorks audits show a course in the wrong requirement. Most are found after students are already affected, which makes catching them at the build step the cheapest fix.

Performance Gap

Actual Performance	Optimal Performance
About 18 section-build error tickets per term (est.)	Section-build error tickets reduced by 50% within two terms, then held at or below 9 per term
0% of department schedule builders have received formal Banner training	100% of active schedule builders complete the module before their next schedule build
About 1 in 4 distance education or hybrid sections need a coding correction (est.)	Fewer than 1 in 20 distance education or hybrid sections need a correction
Schedule builders rely on copying old sections and asking coworkers, and many cannot explain what each field controls	Schedule builders can explain what key fields control and check their own work with a job aid before saving
Cross-listed groups are often set up with caps that do not match the scheduling worksheet	Cross-listed groups match the worksheet on the first build

The gap is mostly a knowledge and skill gap. Schedule builders do not know what the fields control or the rules for coding them, and they have no reliable way to check their work.

Possible Solutions

- Primary (instructional): A 60-minute hands-on module in the Banner test instance that teaches what section fields control, how to build a section, and how to code distance education and cross-listed sections. This is the selected solution.
- Supporting (instructional): A one-page section-build checklist used as a job aid during practice and every live schedule build.
- Supporting (non-instructional): A scheduled query that flags campus, delivery method, and attribute combinations that do not match before registration opens. This catches errors the training misses but does not replace it.
- Considered and not selected: Moving all section building to Registrar staff. It would cut errors but overload a small team during schedule build and would not build skill across departments.

Learner Analysis

The learners are about 25 (est.) department schedule builders and new Registrar scheduling staff who build course sections in Banner each term. They are a subset of the roughly 60 functional Banner users described in Part A.

Number of Learners

The first round will train about 25 learners in four sessions of no more than eight people. After that, the module will run once before each term's schedule build for new hires and anyone who wants a refresher.

Age Group and Gender

Learners are adults between about 24 and 63 years old. Most are women, which matches the makeup of departmental administrative staff at the university. Younger and newer staff are usually open to learning a new process. Long-time staff may be less willing to change a method they have used for years, even if it produces errors.

Socioeconomic Status

Learners are full-time university staff in administrative support and student services roles. Most earn a modest salary, and all have stable employment and university benefits. Training happens during paid work hours, so cost is not a barrier.

Education and Work Experience

Education ranges from a high school diploma to a master's degree, with most holding an associate's or bachelor's degree. Banner experience ranges from a few weeks to more than 15 years. Most schedule builders only build sections a few weeks per term, so their skills fade between schedule builds.

Language Proficiency

All learners read and write English at a professional level. A small number (est. 2 to 3) speak English as a second language. Clear written steps and screenshots on the job aid will support these learners.

Attitude Toward the Subject

Attitudes are mixed. Many schedule builders say they want training because they are tired of having sections sent back for corrections. Others see section building as a small part of their job and may view training as extra work. Some are nervous about "breaking" something in Banner, so practicing in the test environment should lower that anxiety.

Prior Knowledge and Skills

All learners can log in to Banner 9, search for a form, and enter data. Most can copy an existing section and change the term. Few can explain what campus, schedule type, instructional method, and section attributes control in registration, billing, and DegreeWorks. Very few have set up a cross-listed group without help.

Technology Skills

Learners use a computer all day and are comfortable with web browsers, email, and Excel. They are used to Banner's page layout but not to its less common forms, such as SSAXLST.

Impairments and Disabilities

No learners have reported a hearing impairment. One learner (est.) uses reading glasses and prefers larger text, so screens will be zoomed and the job aid will use a large font. No other physical disabilities have been reported. Any accommodation requests will be collected during registration for the session.

Contextual Analysis

Orienting Context

Learners will come in with different goals. Some want to stop having their sections sent back, while others are attending because their supervisor asked them to. Perceived usefulness will be high only if learners see right away that the training fixes problems they deal with each term. To build that, the session will open with real (de-identified) examples of section errors and the student problems they caused. Accountability will come from a hands-on skills check at the end of the session. No skills check exists today; it is a new part of this curriculum. Supervisors will be told who has completed it before the next schedule build.

Instructional Context

- Location: A campus computer lab or training room with one computer per learner. Most learners work in buildings within walking distance.
- Lighting and seating: The room has adjustable lighting and movable chairs. The trainer can dim the lights near the projector so the screen is easy to read.
- Equipment: Each computer has a browser and access to the Banner 9 test instance, a sandbox copy of Banner that is refreshed from live data and holds no real registrations. Learners can make and fix mistakes there without affecting students. The trainer uses a projector to demonstrate each step.

- Access: Learners must already have Banner access with the correct security class for the test instance. Access will be requested at least one week before the session, since missing access is the most likely reason a session could fail.
- Scheduling: Sessions are 60 minutes, held during the work day, and scheduled at least two weeks before schedule build opens. Sessions will not be held during registration or the first week of classes.
- Group size: Capped at eight so the trainer can walk the room and check each learner's screen.
- Noise: The lab is in a quiet office area. Phones will be silenced during the session.

Transfer Context

Learners will use these skills within two to six weeks, during the next schedule build. They will work at their own desks in the live Banner system, often alone and under deadline pressure. To support transfer, each learner will get a one-page section-build checklist and a short reference guide for distance education and cross-listing codes. The Registrar's scheduling staff will review the first sections each learner builds and give quick feedback. Supervisors will be asked to protect time during schedule build so learners are not rushed.

Banner screens and campus coding rules change over time, so transfer also depends on ongoing practice. Trained schedule builders will join a schedule builder community: a shared Teams channel for questions, a short refresher drop-in held before each schedule build, and change notes posted whenever a screen or rule changes. The test instance stays open to learners so they can practice new or rare section types before building them live.

Technology Analysis

Technology	Use in the module	Notes
Banner 9 Student (test instance)	Hands-on practice on SSASECT, SSADETL, and SSAXLST	Practice term and sample courses set up ahead of time so learners do not touch live data
Projector and trainer computer	Trainer demonstrates each step live	Screen zoomed to at least 125% for readability
Lab computers with a web browser	One per learner	Tested the day before to confirm the network reaches the test instance
Printed job aid and scheduling worksheet	Used during practice and kept for the live schedule build	Large font, one page each
Learning management system or shared drive	Post the job aid and a short recorded demo after the session	Supports refresher use before each schedule build

No new software or licenses are needed. The main technology risk is access, so security classes for the test instance will be confirmed before each session.

Unit Goals

At the end of the module, learners will be able to:

1. Explain how campus, schedule type, instructional method, and section attributes affect registration, billing, and DegreeWorks audits. (Understand)
2. Build a complete course section on SSASECT, including meeting time and instructor, that matches the scheduling worksheet with no critical errors. (Apply)
3. Apply the university's coding rules to set up distance education and hybrid sections with the correct campus, instructional method, and attributes. (Apply)
4. Construct a cross-listed group on SSAXLST with enrollment caps that match the scheduling worksheet. (Create)
5. Evaluate a completed section against the section-build checklist and correct any errors before the section is released for registration. (Evaluate)

These goals support the program target from Part A: a 50% drop in section-build error tickets within two terms.

Task Analysis

Topic Analysis

The 60-minute module covers three topics. Each is labeled with its content structure from Chapter 4.

Topic	Content structure	Time
1. What section fields control	Concepts	10 min
2. Building a standard section	Procedure	25 min
3. Coding distance education and cross-listed sections	Principles and rules (with a short procedure)	20 min
Skills check and wrap-up	Assessment	5 min

Topic 1: What section fields control (concepts)

1. Section identity: term, CRN, subject, course number, section number
2. Delivery fields: campus, schedule type (lecture, lab, seminar), instructional method (face-to-face, online, hybrid)

    a. Campus and instructional method drive distance education fees and how the section shows in registration

3. Credit and billing: credit hours, billing hours, grade mode
4. Enrollment fields: maximum enrollment, waitlist
5. Section attributes (SSADETL): used by DegreeWorks and reporting to flag sections such as writing intensive or distance education
6. Section status: active versus cancelled, and when a section is visible to students

Topic 2: Building a standard section (procedure)

1. Read the scheduling worksheet for the section
2. Open SSASECT, enter the term, and create the section from the course
3. Enter section number, campus, schedule type, instructional method, part of term, and status
4. Confirm credit and billing hours match the catalog
5. Enter enrollment limits
6. Add meeting times, building, and room on the Meeting Times and Instructor tab
7. Assign the primary instructor
8. Save, record the CRN, and check the section against the worksheet

Topic 3: Coding distance education and cross-listed sections (principles and rules)

1. Rule: campus, instructional method, and distance education attribute must agree. A fully online section uses the online campus code, the online instructional method, and the distance education attribute.
2. Rule: a hybrid section keeps its main campus but uses the hybrid instructional method and attribute.
3. Rule: add section attributes on SSADETL only when the worksheet calls for them.
4. Procedure: create or join a cross-list group on SSAXLST, add each CRN, and set the group maximum.
5. Rule: the group maximum must equal the room or worksheet cap, and each CRN's maximum must not add up to more than the group maximum.

Procedural Analysis

Topic 1: What section fields control

- What does the learner do? Matches each section field to what it controls in registration, billing, or DegreeWorks. Reviews real error examples and names the field that caused each problem.
- What does the learner need to know? The meaning of each field and which downstream system uses it.
- What cues inform the learner? A problem: the learner cannot explain why a field matters, or picks a value only because last term's section used it. Done: the learner can trace an error example back to the field that caused it.

Topic 2: Building a standard section

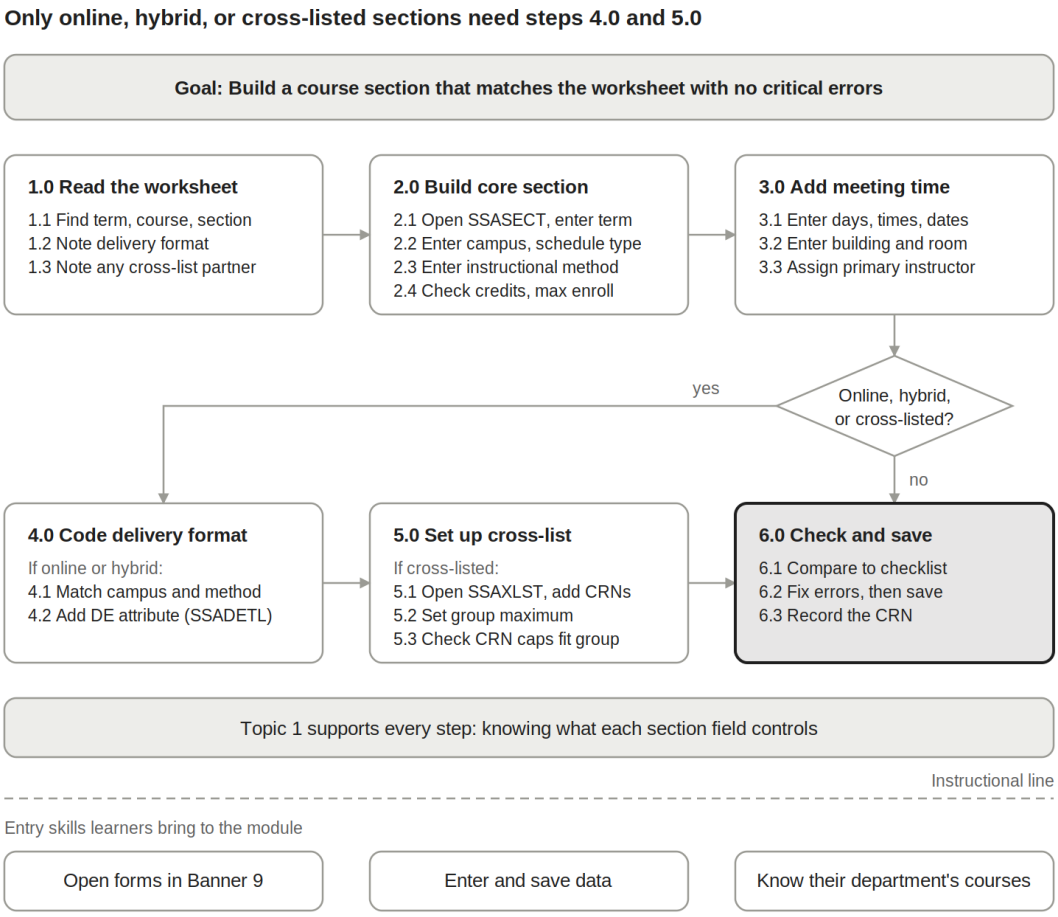
- What does the learner do? Uses the scheduling worksheet to build a new section on SSASECT, adds meeting time and instructor, saves, and checks the section against the worksheet.
- What does the learner need to know? How to move between SSASECT tabs, the valid codes for campus, schedule type, and instructional method, and how to read the scheduling worksheet.
- What cues inform the learner? A problem: Banner shows an error message on save, a field turns red, credit hours do not match the catalog, or a room conflict appears. Done: Banner shows a saved message, a CRN is assigned, and every field matches the worksheet. A different step is needed: if the worksheet shows online or hybrid delivery or a cross-list, the learner moves to Topic 3.

Topic 3: Coding distance education and cross-listed sections

- What does the learner do? Chooses the correct campus, instructional method, and attributes for online and hybrid sections. Adds attributes on SSADETL. Creates a cross-list group on SSAXLST and sets the group maximum.
- What does the learner need to know? The coding rules for each delivery format, how to find SSADETL and SSAXLST, and how group and section maximums relate.
- What cues inform the learner? A problem: the campus and instructional method do not agree, the distance education attribute is missing, or the CRN maximums add up to more than the group maximum. Done: every rule on the checklist is met and the cross-list group shows all CRNs with the correct maximum.

Learning Task Map

The map below shows the steps for Topics 2 and 3, the decision points that send the learner to Topic 3, and the prerequisite skills under the instructional line.



Every section goes through steps 1.0 to 3.0 and 6.0; steps 4.0 and 5.0 apply only when the worksheet shows online or hybrid delivery or a cross-list.

Instructional Objectives

Task	Domain	Instructional objective
Topic 1: What section fields control	Cognitive (concept)	Given six real section error examples, learners will identify the field that caused each error and explain its effect on registration, billing, or DegreeWorks for at least five of the six examples.

Task	Domain	Instructional objective
Topic 2: Building a standard section	Cognitive (procedure)	Given a completed scheduling worksheet and access to the Banner test instance, learners will build a face-to-face section on SSASECT, including meeting time, room, and instructor, within 15 minutes with no critical errors as scored on the section-build checklist.
Topic 3: Coding distance education and cross-listed sections	Cognitive (rule using)	Given scheduling worksheets for one online section and one cross-listed pair, learners will code the campus, instructional method, and SSADETL attributes and build the SSAXLST group so that 100% of coding rules on the checklist are met.

Critical errors are any errors that would change what a student sees in registration, what a student is billed, or how the course counts in DegreeWorks. Examples include a wrong instructional method, a missing distance education attribute, wrong credit hours, or a cross-list cap that does not match the worksheet.

Once all three objectives are met, learners can explain why the fields matter, build a section correctly, and code the two section types that cause the most tickets. This directly addresses the knowledge and skill gap found in the needs analysis. Progress toward the 50% ticket reduction will be checked by comparing section-build tickets for the next two terms against the three-term baseline.